

Sadak Samvad सड़क संवाद

Every citizen's voice + every bus's eyes → **Gemini** → a ranked, funded public-works list for every state.

Live prototype

<https://sadak-samvad-c4c.web.app>

Source

<https://github.com/NaliniSinghRajput/sadak-samvad>

Demo

(demo video link in submission)

Team The Road Rangers

The problem: India's development requests are loud, but fragmented

- Roads, drains and streetlights are what citizens ask for most — via municipal apps, CPGRAMS, WhatsApp, phone calls and ward offices, in **22 scheduled languages**.
- Requests sit in **separate systems**, unverified, and are never compared with **where money is actually sanctioned**.
- Officials see a **queue of complaints**, not a **ranked list of works**; ministries cannot tell whether spending meets demand.
- Citizens who don't write English or use portals are simply **not counted**.

6.3 M km

road network — one of the world's largest, maintained by thousands of agencies

22 + English

languages a national grievance platform must understand — by voice, not just text

0

common layer today linking citizen demand, verified road evidence and public investment plans

Our solution: one Digital Public Good, four steps

1 · Hear everyone

Voice, text, photo · web, **WhatsApp**, IVR, SMS, CSC · any Indian language. **Gemini** transcribes, translates, classifies, grades severity, routes, and **replies in the citizen's language**.



2 · See the road

Camera node on **city buses** detects potholes & hazards on-device, blurs plates, sends events only. **Gemini vision** verifies each crop.



3 · Rank & align

Hotspots scored on demand, severity, camera confirmation, vulnerable users, population (Census). Checked against the **sanctioned-works plan** → unfunded gaps.



4 · Decide

Gemini writes a ministry-ready **policy brief** in any language and answers officials' questions grounded in the data.

Who it serves:

citizens

ward councillors

municipal engineers

State PWD / UDD

MoHUA · MoRTH policymakers

Architecture — built on Google AI and Firebase

Citizen channels

Web/PWA · WhatsApp Business
webhook · IVR voice · SMS ·
Common Service Centres

Bus edge node

4 cameras → one YOLO11 detector
on a laptop-class box · plates
blurred on device · GeoJSON
events (4,308 : 1 less data than
video)

Gemini (multimodal)

- audio + text + photo → structured JSON ticket (schema-constrained)
- vision second opinion on edge crops
- policy brief & grounded Q&A in 23 languages

Priority & alignment engine

~1 km hotspot grid · transparent 5-factor
score · match to sanctioned works (PWD,
Smart Cities, AMRUT, PMGSY)

Firebase

Hosting (global CDN) · Realtime
Database: append-only, validated
public report store

Scale path on Google Cloud

Cloud Run intake API (server-side
key) · BigQuery national analytics ·
Vertex AI hotspot forecasting · Maps
Platform · Speech-to-Text for IVR

Open schemas (GeoJSON events, JSON tickets) · no state-specific code — districts, languages, departments and schemes are configuration.

Google AI is doing the real work

Task	What Gemini does	Why it matters
Multilingual intake	Listens to a voice note / reads text & photo in any of 22 languages → language, transcript, English translation, category (10), severity 1–5, vulnerable users, department, landmark — as schema-constrained JSON	One pipeline replaces manual triage in every language
Citizen reply	Writes the acknowledgement in the citizen's own language & script; spoken aloud by the browser	Voice-first inclusion, trust
Evidence verification	Re-inspects each bus-camera crop, confirms or overrules the edge model, writes the work-order line	Officials act on verified evidence, fewer false alarms
Policy brief	Turns ranked hotspots + spend alignment into a one-page brief with a 7/30/90-day works table, in the reader's language	From data to decision in seconds
Ask the data	Answers officials' questions grounded only in the platform's numbers	No dashboards skills needed

Edge computer vision (YOLO11 + our pothole model, test mAP50 0.51; plate detector mAP50 0.86) runs on the bus; Gemini is the cloud intelligence on top.

Working prototype — citizen intake

Sadak Samvad ସଡ଼କ ସଂବାଦ

Citizen voice + bus-camera evidence → Gemini → ranked public works, for every state

1 · Citizen report 2 · Governance dashboard 3 · Bus-camera evidence 4 · Policy brief & Ask About

Gemini live · gemini-3.6-flash

Report a road or civic problem — any language, voice, text or photo

District: Khordha (Bhubaneswar) — Odisha Channel: Web / app

What is the problem? (type in Hindi, Odia, Tamil, Bengali... or English)

ରସୁଲଗର ଛକ ପାଖରେ ରାସ୍ତା ଭାଙ୍ଗିଯାଇଛି, ବର୍ଷାରେ ପାଣି ଜମି ରହୁଛି ଓ ବାଇକ୍ ଖସିପଡୁଛି।

Speak (record) Add photo Use my location

Location: district centre (tap the map or use your location to refine)

Map showing Bhubaneswar and surrounding areas.

What Gemini understood

Language	Odia
Heard / read	ରସୁଲଗର ଛକ ପାଖରେ ରାସ୍ତା ଭାଙ୍ଗିଯାଇଛି, ବର୍ଷାରେ ପାଣି ଜମି ରହୁଛି ଓ ବାଇକ୍ ଖସିପଡୁଛି।
English	The road near Rasulgarh Square is broken, water accumulates during rains, and bikes are skidding/falling.
Category	Pothole / road surface
Severity	4 / 5 vulnerable users affected
Routed to	PWD / Municipal roads
For the official	Damaged road surface and potholes near Rasulgarh Square causing water accumulation and two-wheeler accidents.
Photo	The photo shows an uneven, damaged asphalt road surface with dirt patches and pothole depressions near a busy street.
Landmark	Rasulgarh Square
Ticket	LV-muodbpju · added to this session · gemini-3.6-flash

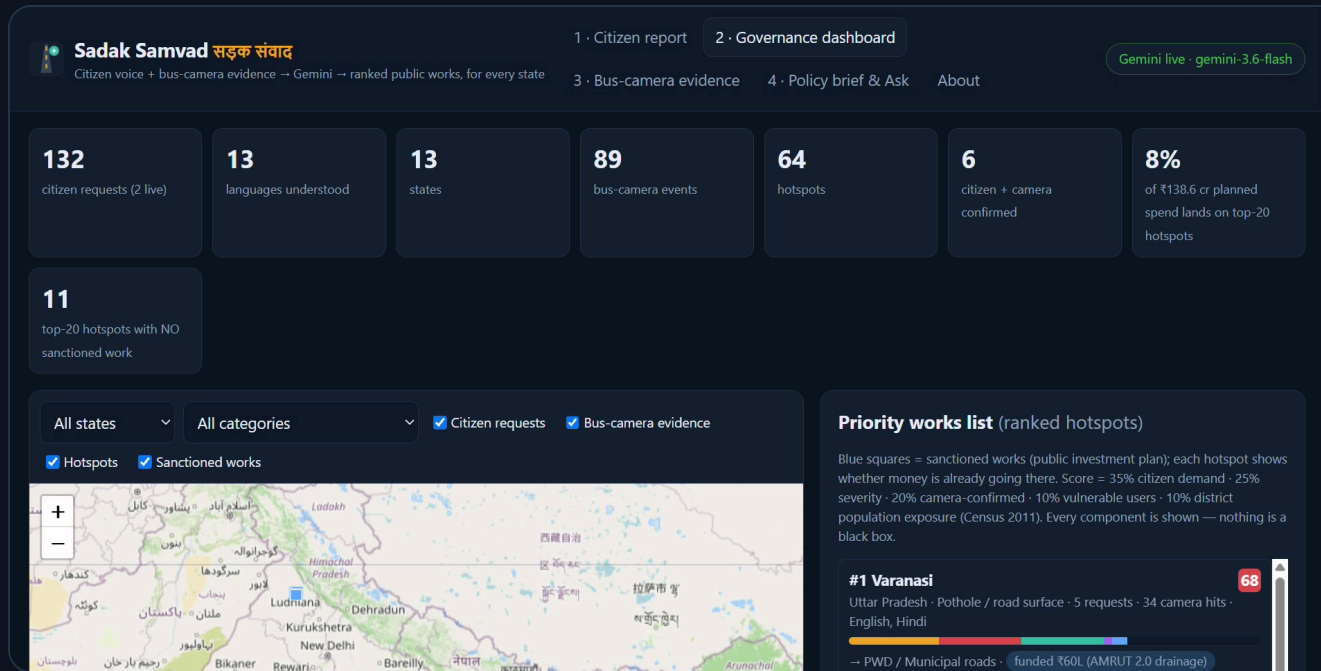
ଆପଣଙ୍କ ଅଭିଯୋଗ ପଢ଼ିବୁନି ହୋଇଛି। ଏହା ଶୀଘ୍ର ସମାଧାନ ପାଇଁ PWD / Municipal roads ବିଭାଗକୁ ପଠାଯାଇଛି।

Recent reports on this district

- Pick district & channel; **type, speak or snap a photo**
- Gemini returns a structured ticket in seconds
- Reply read back in the citizen's language
- Ticket is stored live in **Firestore Realtime Database** and appears on every official's map
- Try it: Hindi, Odia, Tamil, Bengali examples built in

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Governance dashboard — hotspots, priorities, spending gaps



- 130+ requests, 13 languages, 14 districts, 13 states in the sample
- Transparent score: 35 % demand · 25 % severity · 20 % camera · 10 % vulnerable · 10 % population
- Every hotspot marked **funded** (which scheme, ₹) or **UNFUNDED GAP**
- KPI: share of planned spend landing on top-20 hotspots — the misalignment the track asks us to expose

Real evidence from real buses — not a mock-up

Every defect, on the road we actually drove

KIIT → Nexus Esplanade, Bhubaneswar · 9.3 km surveyed of the 13.5 km route · morning drive, 19 Sep 2026



FOUND ON ONE PASS OF THIS ROUTE

48 potholes, each graded for severity

141 further road-damage candidates

3,944 detections in all, across four cameras

SEVERITY OF THE 48 POTHOLE

● 18 severe ● 20 moderate ● 10 minor

A repair crew gets 18 addresses to fix first, in order — not a queue of complaints.

The route is the one driven. Each defect is placed along it by the time it was detected — interpolated, not a per-event satellite fix. Given a position feed the node geo-tags from real GPS only, and refuses to place an event without one.

3,944

detections, 4-camera drive, Bhubaneswar, 19 Sep 2026

48

potholes graded (18 severe) on one pass

34

GNSS-tagged potholes, Varanasi, 29 Aug 2026

4,308:1

video in : data out — events, not video

One laptop-class box serves 4 cameras; plates blurred before anything leaves the bus. Gemini vision then verifies each crop.

Impact — who benefits, and how much

Citizens

Anyone with a phone — in their own language, by voice — gets a ticket, a department and a reply. Includes people who never use portals (via IVR/CSC).

Cities & states

Engineers get a verified, ranked works list instead of a complaint queue; buses already on the road become free survey vehicles every day.

Policymakers

For the first time, a measure of whether sanctioned money meets citizen demand — district by district, state by state.

4,000+

statutory towns & cities that can onboard with configuration only

22

scheduled languages + English understood by one model

Daily

road survey from routine bus runs — no survey vans

Safety

potholes & waterlogging fixed by risk, vulnerable users weighted

Deployable in weeks — a pilot plan a ministry can sign

Week 1–2

One Municipal Corporation: load district, ward & department config; connect grievance export and works register; WhatsApp number live.

Week 3–4

Fit edge nodes on 10 city buses (phones + one laptop-class box each); events flow to the dashboard.

Week 5–6

Engineers work from the ranked list; Gemini briefs to Commissioner weekly; measure time-to-fix.

Week 8

State PWD / UDD dashboard across all ULBs; feed state works register; go to next state.

Low cost

Static hosting + serverless + Gemini Flash; reuses existing buses and phones.

Privacy by design

No video leaves the bus; plates blurred on device; citizen data minimal; append-only validated store.

Open & interoperable

Open source, open schemas; plugs into CPGRAMS/state portals; DPG-ready.

Next phase (in progress): BigQuery + Vertex AI

BigQuery — the national data layer

- Firebase → BigQuery streaming export of every citizen ticket and bus-camera event (lakhs/day, 4,000+ towns).
- Joined with Census, data.gov.in, PMGSY / Smart Cities / AMRUT works registers and budget releases.
- Hotspot score computed in SQL for every ward of every state, nightly.
- **"% of sanctioned spend landing on citizen hotspots"** per district, state and scheme — measuring the impact of DPI spending for MoHUA / MoRTH.

Vertex AI — prediction & prioritisation

- **Forecasting:** next month's pothole and monsoon-waterlogging hotspots per ward from tickets + camera events + IMD rainfall — fix before the rains.
- **Custom training & serving:** retrain the pothole detector on the growing bus-camera dataset; push better models to every bus edge box.
- **Prioritisation model:** learn from which works were fixed and how fast; recommend order, cost and time.
- **Gemini on Vertex AI:** enterprise quotas, India data residency, server-side keys.

Citizens + buses → **Gemini** (understand) → **Firebase** (live) → **BigQuery** (national join & metrics) → **Vertex AI** (forecast & prioritise) → **Gemini** brief to policymakers

Scales by configuration, not code — a new state = district list + languages + departments + works register; open schemas let any BRICS city fork it with its own languages.

Honest by design · credits · links

Data used

- **Real:** bus-camera detections from our own drives (Varanasi 29 Aug; Bhubaneswar 19 Sep 2026); Census 2011 district populations.
- **Synthetic, labelled "sample":** 130 citizen requests and 61 sanctioned works — stand-ins until CPGRAMS/state feeds are connected. Every report entered live is real.

What is new for this hackathon

The edge node began as our SIH 2026 work (PS 26124). New here: Gemini multilingual intake, Gemini verification, fusion & priority engine, investment alignment, Gemini policy layer, Firebase live store, the whole governance app.

Built with

Gemini API · Firebase Hosting & Realtime Database · Leaflet (BSD-2) · OpenStreetMap (ODbL) · marked (MIT) · Ultralytics YOLO11 (AGPL-3.0) · OpenCV (Apache-2.0) · FastAPI (MIT) · Roboflow pothole dataset (CC BY 4.0)



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सड़क संवाद — every voice counted, every rupee aligned.